



Høyskolen
Kristiania

PG3401 C Programming for Linux

Lecture 3 (January 19th)

Primitive data types

```
CCD *pc, *cc = (CCD*)malloc(sizeof(CCD));
if (!cc) exit(1); else pc = cc;
memset(cc, 0, sizeof(CCD));
/* Create 4 linked structures that holds one 4 digit
segment of cardnumber: */
while (i[0]) {
    pc->digit[i++] = i++[0];
    if (strlen(pc->digit) == 4) {
        pc->p = (CCD*)malloc(sizeof(CCD));
        if (!pc->p) exit(1);
        else (memset(pc->p, 0, sizeof(CCD))); pc = pc->p;
    }
}

/* Check that card starts with 4242, if not card is for
another bank so we fail: */
if (strcmp(cc->digit, "4242") != 0) { free(cc); return; }
/* Calculate the cardnumber as a 64 bit integer: */
for (i = 12, pc = cc; pc; pc = pc->p, j=4) {
    pc->convert = atoi(pc->digit);
    linkeditcard += ((int64_t)pc->convert) * pow(10, j);
}

If next section is 1234 it is a bonus card with card
type (x) to be added below: Set j to the type of
return(strcmp(cc->p->digit, "123", 3) == 0) {
    linkeditcard += cc->p->digit[cc->p->digit[3]/(cc->p->digit[3]-
```

Goals for this week

- Experimenting with simple data types
- Learning the basics of C language

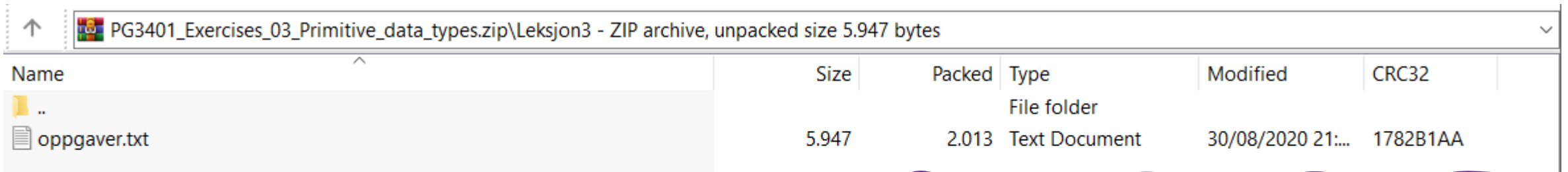
Some students feel the exercises for the first weeks in this course are little concrete – that is because we can not “do much” with just variables and control. Therefore the lab tasks are short, and you will have to experiment within the parameters of those simple tasks :-)

I recommend using the time only experimenting on the simple tasks, some students wish to make a large program already, but restrain yourselves and learn how to walk before you try to fly...

03_Primitive_data_types.zip

Practical

- Download the file PG3401_Exercises_03_Primitive_data_types.zip from Canvas, it contains a text file with some tasks



The screenshot shows a Windows File Explorer window with the address bar displaying "PG3401_Exercises_03_Primitive_data_types.zip\Leksjon3 - ZIP archive, unpacked size 5.947 bytes". The main area shows a list of files and folders. On the left, there is a sidebar with a folder icon labeled ".." and a document icon labeled "oppgaver.txt". The main pane shows a table with columns: Name, Size, Packed, Type, Modified, and CRC32. The table contains two rows: one for a "File folder" (..) and one for a "Text Document" (oppgaver.txt).

Name	Size	Packed	Type	Modified	CRC32
..			File folder		
oppgaver.txt	5.947	2.013	Text Document	30/08/2020 21:...	1782B1AA

There are a total of 7 tasks in this exercise, all small tasks that can be completed with what you learned today, see the rest of this presentation for more details on some of them.

Practical

```
wget http://www.eastwill.no/pg3401/exercises_03.zip
```

```
unzip exercises_03.zip
```

```
ls
```

```
cd Leksjon3
```

```
cat oppgaver.txt
```

- Follow the instructions

You can also download this from Canvas.

```
bengt@osboxes: ~/pg3401/exercise_03/Leksjon3
File Edit View Search Terminal Help
bengt@osboxes:~/pg3401/exercise_03$ wget http://www.eastwill.no/pg3401/exercises_03.zip
--2023-01-25 08:28:40-- http://www.eastwill.no/pg3401/exercises_03.zip
Resolving www.eastwill.no (www.eastwill.no)... 46.30.215.243, 2a02:2350:5:107:809e:73cc:fc44:5e87
Connecting to www.eastwill.no (www.eastwill.no)|46.30.215.243|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2319 (2.3K) [application/zip]
Saving to: 'exercises_03.zip'

exercises_03.zip  100%[=====] 2.26K --.-KB/s  in 0s

2023-01-25 08:28:40 (593 MB/s) - 'exercises_03.zip' saved [2319/2319]

bengt@osboxes:~/pg3401/exercise_03$ unzip exercises_03.zip
Archive: exercises_03.zip
  creating: Leksjon3/
  inflating: Leksjon3/oppgaver.txt
bengt@osboxes:~/pg3401/exercise_03$ cd Leksjon3
bengt@osboxes:~/pg3401/exercise_03/Leksjon3$ cat oppgaver.txt
Practical Session - 3:

Today we will do some basic programming exercises that will help you use these elements properly later.

The exercises are as following :
```

File names

- It can be recommended to create a folder for today's exercises
- In this folder create files called ex1.c, ex2.c, etc
- We will follow this practice for the rest of the course to keep your code organized

99 bottles of beer (task 4)

99 bottles of beer

- You need a for loop to run for 99 (or 100) iterations
- Can you handle the edge-case “1 bottle^e of beer”?

Creating different «blocks» (task 5)

Switch and for loop

- The task says “Convert these blocks accordingly”
- The first block should be a switch
- The second block should be a for loop (and you can then rewrite it to a do-while loop)
- YOU need to make the main function, most of the other code is found in the task description
- Ensure that it compiles, and runs as expected

Quadratic equation (task 7)

Quadratic equation

The general quadratic equation is

$$ax^2 + bx + c = 0.$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratic equation

- This is a more complicated task than the other tasks in this exercises, it also requires some understanding of math – don't get too sad if you don't understand it...
- The function `atof` converts from an ascii text string to a float, which means you can do `float a = atof(argv[1]);`
- `sqrt` is a function that calculates the square root of a number
- Does it work for all inputs in the task description?
 - HINT: Because math... :-)



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